STUDENT ENROLLMENT COURSE & MANAGEMENT SYSTEM (SECMS)

2. REQUIREMENT ANALYSIS

2.1 Project Overview

The Student Enrollment & Course Management System (SECMS) is a Salesforce-based centralized system designed to manage student registration, courses, enrollments, instructors, approvals, payments, and reporting.

3.1 Customer Journey Map

Stages Covered:

- Awareness & Registration
- Course Selection
- Enrollment Request

Fee Payment

- Approval
- Instructor Assignment

Reporting & Completion

- Customer Journey Map – SECMS
- Awareness & Registration

Think/Feel:

- Concerned about multiple logins
- Worried about revenue loss
- Course Selection
- Enrollment Request

Hear:

- Complaints from students

- Feedback from management

Fee Payment

- Approval
- Instructor Assignment

Say/Do:

- Automation, dashboards
- Faster processing

2.2 Functional Requirements

ID	Requirement	Description
FR-1	**Student Management**	**Manage student record**
FR-2	**Course Management**	**Create and manage courses**
FR-3	**Enrollment Processing**	**Request, approve, reject enrollments**
FR-4	**Instructor Assignment**	**Auto-assign instructors**
FR-5	**Reporting**	**Dashboards and reports**

2.3 Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

Examples:

Quick student registration form

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

Examples:

Only admins can modify course details

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

Examples:

No duplicate course registration

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples:

Quick dashboard refresh

Fast search for student records

No lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

Examples:

Add new departments without redesign

Handle increased admissions annually

Extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

Examples:

24/7 cloud access

Data Flow Diagram

Data Flow

User inputs student and enrollment data

System validates and stores records

Approval process routes data

Reports and dashboards consume processed data

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

DFD Flow

Student/User

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Enrollment Data

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Validate & Store Records

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SECMS Database

2.5 Technology Stack

- Layer Technology
- UI. Salesforce Lightning
- Logic Salesforce Flows
- Database Salesforce Objects
- Automation

- Validation Rules, Approval Process
- Reporting
- Reports & Dashboards

2.6 Main Data Entities

The main data entities in the Student Enrollment & Course Management System (SECMS) are Student, Course, Enrollment, and Instructor. The Student entity stores student details, while the Course entity manages course information. The Enrollment entity records student enrollment requests and their status, and the Instructor entity maintains instructor details and supports instructor assignment. These entities work together to manage the complete enrollment process.

2.7 Requirement Summary

The requirements establish a centralized, secure and cloud-based system with structured data management, validation, relationships and analytics. These requirements guide the design and implementation phases.